Ramanujan, Landen, and Baker N-Series Refinement: Small-Nome Gap Asymptotics and a Two-Axis Comparison

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Abstract

We compare three distinct refinement mechanisms: Ramanujan's ellipse-perimeter approximation along the eccentricity axis, fixed-scale geometric N-series cancellation along the resolution axis, and Landen–arithmetic–geometric mean (AGM) contraction along the modulus axis. The classical theta-function identity $\delta(q) = k(q^2)$, for the normalized complementary-modulus gap $\delta(q) = (1 - k'(q))/(1 + k'(q))$, directly yields, for every fixed integer $m \geq 2$,

$$\frac{\delta(q^m)}{\delta(q)^m} = 4^{1-m}\left(1 + 4mq^2 + O(q^4)\right), \qquad q \to 0.$$

The first-correction coefficient is thus linear in m; this is a small-nome asymptotic, not a newly derived degree-m modular equation. With explicit Gauss–Legendre indexing, classical AGM error asymptotics imply $\alpha_n/\delta_n^2 \to 1/8$ and $\alpha_{n+1}/\alpha_n^2 \to 1/2$. We separate these specific coefficients from generic normalized quadratic and cubic error maps. We make no claim that geometric N-series extrapolation is a modular transformation.

1 Ramanujan's approximation: the eccentricity axis

For ellipse semiaxes $a \geq b > 0$, set $h = ((a-b)/(a+b))^2$. The exact perimeter is

$$P = \pi(a+b)\,{}_2F_1(-\tfrac{1}{2}, -\tfrac{1}{2}; 1; h) = \pi(a+b)\left(1 + \tfrac{1}{4}h + \tfrac{1}{64}h^2 + \tfrac{1}{256}h^3 + \tfrac{25}{16384}h^4 + \cdots\right).$$

Ramanujan's second approximation is

$$R_2 = \pi(a+b)\left(1 + \frac{3h}{10 + \sqrt{4 - 3h}}\right).$$

It agrees through h^4, and our *signed-error convention* is

$$\boxed{P - R_2 = \pi(a+b)\left(\frac{3}{131072}h^5 + O(h^6)\right).}$$

Equivalently, the coefficient in $R_2 - P$ is $-3/131072$. This mechanism is an eccentricity expansion; it does not refine a polygonal resolution variable.

2 Geometric N-series: the resolution axis

If a polygonal perimeter family admits an even-power expansion of suitable order,

$$P_N = P + C_2 N^{-2} + C_4 N^{-4} + C_6 N^{-6} + \cdots,$$

Richardson-type linear combinations of $P_N, P_{rN}, \ldots$ may cancel selected terms. Cancelling the first s error terms yields $O(N^{-2(s+1)})$ provided the necessary expansion and remainder bound hold. The cancellation coefficients arise from a finite Vandermonde/moment system. This algebraic convergence in N is distinct from nome-squaring under the descending Landen transformation and from the cubic trisection residual. No fixed number of Richardson columns creates a modular transformation.

3 Landen contraction and explicit AGM indexing

Define the Gauss–Legendre iteration by

$$\begin{array}{llll} a_0 = 1, & b_0 = 2^{-1/2}, & t_0 = \frac{1}{4}, & p_0 = 1, \\ a_{n+1} = \frac{1}{2}(a_n + b_n), & b_{n+1} = \sqrt{a_n b_n}, & t_{n+1} = t_n - p_n(a_n - a_{n+1})^2, & p_{n+1} = 2p_n. \end{array}$$

We use *matching indices* throughout:

$$\boxed{\pi_n = \frac{(a_n + b_n)^2}{4t_n}, \quad \delta_n = \frac{a_n - b_n}{a_n + b_n}, \quad e_n = \pi_n - \pi, \quad \alpha_n = -\frac{e_{n+1}}{e_{n+1} - e_n}.}$$

In particular π_n is formed from (a_n, b_n, t_n), not from the next iteration's data. Altering this convention alters intermediate identities and may alter reported constants. The AGM update gives the exact descending Landen relation

$$\delta_{n+1} = \frac{1 - \sqrt{1 - \delta_n^2}}{1 + \sqrt{1 - \delta_n^2}} = \frac{\delta_n^2}{(1 + \sqrt{1 - \delta_n^2})^2} = \frac{1}{4}\delta_n^2 + O(\delta_n^4).$$

For the above Gauss–Legendre convention, the classical error asymptotics can be expressed as

$$e_n \sim -\pi^2 2^n \delta_n^2.$$

Consequently $|e_{n+1}|/|e_n| \to 0$, and the exact definition of α_n gives

$$\alpha_n \sim \frac{e_{n+1}}{e_n} \sim 2\frac{\delta_{n+1}^2}{\delta_n^2} \sim \frac{1}{8}\delta_n^2, \qquad \boxed{\lim_{n\to\infty} \frac{\alpha_n}{\delta_n^2} = \frac{1}{8}.}$$

The same relations imply

$$\boxed{\lim_{n\to\infty} \frac{\alpha_{n+1}}{\alpha_n^2} = \frac{1}{2}.}$$

These limiting constants use the stated definitions; they are not invariant under arbitrary shifts in the approximate-π indexing.

4 Small-nome gap asymptotics, not higher-degree modular equations

Let $q = e^{i\pi\tau}$, $\operatorname{Im}\tau > 0$, and take $0 < q < 1$ in the real-valued statements. Write

$$k(q) = \frac{\vartheta_2(q)^2}{\vartheta_3(q)^2}, \quad k'(q) = \sqrt{1-k(q)^2}, \quad \delta(q) = \frac{1-k'(q)}{1+k'(q)}.$$

The classical quadratic theta identities give the exact identity

$$\boxed{\delta(q) = k(q^2).} \tag{1}$$

Indeed $\vartheta_3(q)^2 + \vartheta_4(q)^2 = 2\vartheta_3(q^2)^2$ and $\vartheta_3(q)^2 - \vartheta_4(q)^2 = 2\vartheta_2(q^2)^2$; since $k'(q) = \vartheta_4(q)^2/\vartheta_3(q)^2$, their quotient yields (1). The classical theta product then yields

$$k(Q) = 4\sqrt{Q}\prod_{j=1}^{\infty}\left(\frac{1+Q^{2j}}{1+Q^{2j-1}}\right)^4, \quad \delta(q) = k(q^2) = 4q\left(1-4q^2+O(q^4)\right).$$

Proposition 1 (Fixed-degree small-nome gap ratio)**.** *For every* fixed *integer $m \geq 2$, as $q \to 0^+$,*

$$\boxed{\frac{\delta(q^m)}{\delta(q)^m} = 4^{1-m}\left(1+4mq^2+O(q^4)\right).}$$

In particular, the ratio tends to 4^{1-m}.

Proof. By (1), $\delta(q^m) = k(q^{2m}) = 4q^m(1-4q^{2m}+O(q^{4m}))$ and $\delta(q)^m = 4^m q^m(1-4q^2+O(q^4))^m$. Divide, expand the inverse denominator, and note that $2m \geq 4$. The coefficient of q^2 is therefore $4m$. □

Remark 1 (Terminology and priority)**.** *This statement concerns the leading and first-correction asymptotics of $k(q^{2m})$ relative to $k(q^2)^m$. It is* not *the classical degree-m modular polynomial or a new modular equation. The ingredients are standard theta-function identities and products; no historical priority is claimed for them.*

5 Numerical illustration

For $R_m(q) = \delta(q^m)/\delta(q)^m$, define $C_m(q) = q^{-2}(R_m(q)/4^{1-m} - 1)$. The proposition predicts $C_m(q) \to 4m$. At $q = 10^{-12}$ with 400-digit working arithmetic, the reported values are:

m	$4m$	$C_m(10^{-12})$
2	8	8.000000000000000000000016
3	12	12.000000000000000000000054
4	16	16.000000000000000000000104
5	20	20.000000000000000000000170
6	24	24.000000000000000000000252

These are numerical illustrations, not the proof. To avoid cancellation when q is tiny, use $\delta(q) = \lambda(q)/(1+\sqrt{1-\lambda(q)})^2$, where $\lambda(q) = k(q)^2$. A canonical repository command is

```
python scripts/verify_modular_gap_law.py --precision 400 --outdir outputs
```

6 What normalization does and does not show

For a local error law $x_{n+1} = \lambda x_n^p + O(x_n^{p+2})$ with $\lambda \neq 0$, constant rescaling can normalize the leading coefficient to ± 1; hence the bare normalized form primarily encodes the local *order* p. In particular, the Landen law is $\delta_{n+1} = \frac{1}{4}\delta_n^2 + O(\delta_n^4)$, while with $\eta_n = \delta_n/4$ it becomes $\eta_{n+1} = \eta_n^2 + O(\eta_n^4)$. These equations encode the same quadratic contraction.

The companion Baker 4:3 trisection analysis records a cubic residual and, in its specified geometric normalization, the coefficient $-7/384$. Where its signed error variable x_n obeys $x_{n+1} = -(7/384)x_n^3 + O(x_n^5)$, putting $\eta_n = \sqrt{7/384}\, x_n$ gives $\eta_{n+1} = -\eta_n^3 + O(\eta_n^5)$. The value $7/384$ belongs to the *chosen geometric residual*, whereas the normalized equation by itself cannot recover that value. The comparison between degree two and degree three is therefore a comparison of convergence orders, not a claim of a shared special coefficient or of modular equivalence.

AI-assistance and review disclosure

ChatGPT and Claude were used for drafting assistance, mathematical checking, numerical diagnostics, and editorial review. High-precision agreement is computational evidence, not independent human peer review. The analytical claims remain subject to the stated hypotheses and cited classical results.

7 Conclusion and limits

Ramanujan's second ellipse approximation achieves unusually high-order cancellation in the eccentricity parameter h; the geometric N-series removes specified even powers in polygonal resolution N^{-1}; and the AGM/Landen iteration contracts a modulus-like gap quadratically. The exact theta identity $\delta(q) = k(q^2)$ streamlines the small-nome ratio proof, clarifies the nomenclature, and situates the result within classical modular theory. The one-eighth AGM constant is tied to explicitly indexed iterates, and the cubic trisection coefficient cannot be inferred from a normal form alone. These mechanisms are comparable as forms of error suppression, but are not mathematically identical.

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